Ex. No: 4

Title: Entropy, Cross-Entropy and KL Divergence

Packages: NumPy, SciPy, Matplotlib

AIM:

To calculate entropy, cross-entropy and KL divergence for two probability distributions.

ALGORITHM:

1. Start the program and import NumPy, SciPy and Matplotlib.
2. Take the true distribution P and the predicted distribution Q.
3. Calculate entropy $H(P) = -\Sigma\ P \log_2 P$.
4. Calculate cross-entropy $H(P,Q) = -\Sigma\ P \log_2 Q$.
5. Calculate KL divergence $D(P||Q) = \Sigma\ P \log_2 (P/Q)$.
6. Plot the three values as a bar chart.
7. Display the output and stop.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import entropy

P = np.array([0.4, 0.3, 0.2, 0.1])
Q = np.array([0.25, 0.25, 0.25, 0.25])

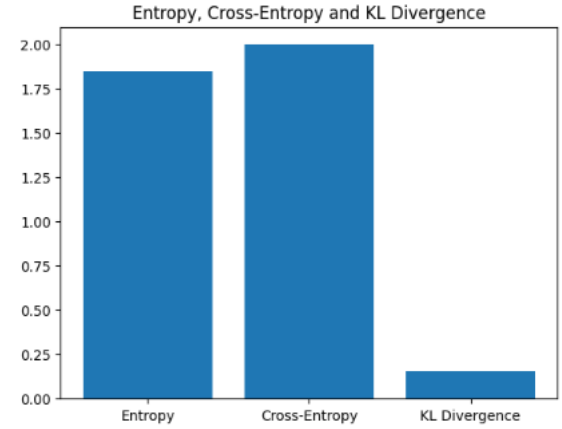
H = entropy(P, base=2)
KL = entropy(P, Q, base=2)
CE = -np.sum(P * np.log2(Q))

print("Entropy H(P) =", round(H, 4))
print("Cross-Entropy H(P,Q) =", round(CE, 4))
print("KL Divergence =", round(KL, 4))

plt.bar(["Entropy", "Cross-Entropy", "KL Divergence"], [H, CE, KL])
plt.title("Entropy, Cross-Entropy and KL Divergence")
plt.show()
```

OUTPUT:

```
Entropy H(P) = 1.8464
Cross-Entropy H(P,Q) = 2.0
KL Divergence = 0.1536
```



RESULT:

Entropy, cross-entropy and KL divergence were calculated successfully. Entropy = 1.8464, Cross-Entropy = 2.0 and KL Divergence = 0.1536.